

Prove that the VC-dimension of $\mathcal{F}$ is $d + 1$, where:

$$\mathcal{F} = \{\text{sgn}(\mathbf{w}^T \mathbf{x} + b) \mid \mathbf{w} \in \mathbb{R}^d, b \in \mathbb{R}\}$$

where:

- $\mathbf{w} \in \mathbb{R}^d$ is the weight vector
- $b \in \mathbb{R}$ is the bias term
- $\mathbf{x} \in \mathbb{R}^d$ is the input vector
- $\text{sgn}(\cdot)$ is the sign function: $\text{sgn}(z) = \begin{cases} +1 & \text{if } z > 0 \\ -1 & \text{if } z \leq 0 \end{cases}$

- 先证明 $VCdim(\mathcal{F}) \geq d + 1$:

选 $d+1$ 个点:

$$x_1 = (0, 0, \dots, 0)^T, x_2 = (1, 0, \dots, 0)^T, x_3 = (0, 1, \dots, 0)^T, \dots, x_{d+1} = (0, 0, \dots, 1)^T$$

任取标签向量:

$$y = (y_1, \dots, y_{d+1})^T \in \{-1, 1\}^{d+1}$$

我们希望:

$$\text{sgn}(w^T x_i + b) = y_i, \quad \forall i$$

为方便表示, 取:

$$z_i^T = (x_i^T, 1), \\ \tilde{w}^T = (w^T, b)$$

从而有:

$$\text{sgn}(\tilde{w}^T z_i) = y_i, \\ y_i(\tilde{w}^T z_i) = 1$$

只需要取:

$$Z = (z_1, z_2, \dots, z_{d+1}), \\ \tilde{w} = Z^{-T} y$$

即有:

$$\begin{aligned} y_i(\tilde{w}^T z_i) &= y_i((y^T Z^{-1}) z_i) \\ &= y_i(y^T (Z^{-1} z_i)) \\ &= y_i(y^T e_i) \\ &= y_i \cdot y_i \\ &= 1 \end{aligned}$$

其中:

$$Z e_i = z_i, \quad \text{therefore } Z^{-1} z_i = e_i$$

从而实现了期望的分类, 因此证明了 $VCdim(\mathcal{F}) \geq d + 1$.

- 再证明 $VCdim(\mathcal{F}) \leq d + 1$:

任取 $d + 2$ 个点 $x_1, x_2, \dots, x_{d+2}$, 根据Radon's theorem, 存在其划分A, B使得:

$$conv(A) \cap conv(B) \neq \emptyset$$

因此只需给A中全部标记为+1, B中全部标记为-1, 若存在 w, b 能实现分类, 则 $conv(A), conv(B)$ 在超平面的异侧, 不应相交, 矛盾。

从而无法实现期望分类, 因此证明了 $VCdim(\mathcal{F}) \leq d + 1$.

综上有:

$$VCdim(\mathcal{F}) = d + 1$$